GX Backup — Recovery and Reinstall Guide

Dual ASUS GX10 cluster (gx10-01 + gx10-02). Generated 2026-09-23.

EMERGENCY RECOVERY

If a machine dies: (1) On the surviving node run **gx doctor**. (2) Install Ubuntu 24.04 on the replacement GX10. (3) Clone the private **legenex/gx-backup** repository to `/home/legenex/Documents/Projects/gx-backup`. (4) Copy `~/.config/gx-backup/restic-password` and `age.key` from the surviving node. (5) Run `./install` then `gx restore latest`. (6) Recreate model weights from `models.lock`. (7) Start compose + user systemd units. (8) **gx doctor**.

What GX Backup is

Git holds the rebuild recipe (scripts, pins, sanitized configs, this guide). Restic holds encrypted mutable data. Public model weights are not in Git and are not duplicated into every snapshot.

Everyday commands

```
gx status
gx backup
gx doctor
gx models
gx auto
gx mini
gx code
gx max
gx logs
gx benchmark
```

What	Where
Recipe	GitHub legenex/gx-backup (private) and this checkout
Encrypted data gx10-01	~/Documents/Backups/restic/local and peer copy on gx10-02
Encrypted data gx10-02	~/Documents/Backups/restic/local on gx10-02 and peer copy on gx10-01
Restic password	~/.config/gx-backup/restic-password on both nodes (0600). Never in Git.
Models	/srv/models plus models.lock pins and sha256
Gateway	http://127.0.0.1:4000/v1 models gx-auto gx-mini gx-code gx-max
OpenWebUI	http://127.0.0.1:3000 volume open-webui
GX-Cluster	http://127.0.0.1:8088 Backup page starts gx backup

Restore one file / project / database

```
export RESTIC_PASSWORD_FILE=$HOME/.config/gx-backup/restic-password
restic -r $HOME/Documents/Backups/restic/local snapshots
```

```
restic -r $HOME/Documents/Backups/restic/local restore latest \
  --target /tmp/restore --include /path/to/file
# SQL dumps: Documents/Backups/dumps/<stamp>/*.sql.gz
# OpenWebUI: open-webui-data.tar.gz into the docker volume
```

Rebuild from fresh Ubuntu

Same hardware class (ASUS GX10 / NVIDIA GB10). Create the matching user, install Docker CE, add the user to docker. Configure Tailscale and the two ConnectX-7 nets (192.168.100.10/11 and 192.168.101.10/11). SSH host gx10-02 uses key id_ed25519_gxcluster. Then follow Emergency Recovery above. Do not reboot both nodes at once.

Nodes discover each other

SSH: ~/.ssh/config Host gx10-02 → 100.73.238.4 user legenex-02. Model traffic: RoCE 192.168.100.11:28080, never Tailscale. Management: Tailscale 100.105.214.61 (gx10-01) and 100.73.238.4 (gx10-02).

Common failures

gx-mini exit 137: raise cgroup memory (24g) and keep ctx at 65536. Empty restic password: export RESTIC_PASSWORD_FILE. LiteLLM 401: missing gateway key. Orchestrator only on loopback: restart after docker0 has an address.

Updating this recipe

Edit files in gx-backup, run scripts/secret-scan.py, commit, push the private repo. Never commit .env, restic-password, age.key, or SSH private keys.